

Project Design Phase
Problem – Solution Fit Template

Date	29 September 2026
Team ID	SWTID-2026-5115
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer’s problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- To restrict access to institution records based on user roles and branch-specific conditions in the ServiceNow platform.
- To ensure that only authorized users can view, create, modify, or delete records according to their assigned permissions.
- To protect sensitive student information and prevent unauthorized users from accessing records belonging to other branches.
- To implement Script-Controlled Access Control Lists (ACLs) for effective role-based access management.
- To allow authorized EEE users to access EEE branch records while ensuring administrators retain full access.
- To improve data security and privacy by validating user roles and record conditions before granting access.
- To help administrators manage user permissions effectively and reduce the risk of unauthorized record operations.
- To provide a secure and reliable method for managing institution records in ServiceNow.

1. CUSTOMER SEGMENT(S) CS • Institution / ServiceNow administrators who manage student and institution records. • Staff and faculty users who hold roles bb1, bb2, bb3 and bb4. • EEE branch users who should view only EEE student records.	6. CUSTOMER CONSTRAINTS CC • Solution must run only on the ServiceNow platform (roles, ACLs, scripting); no external tools or APIs. • Users cannot get access without the correct role assignment by an administrator. • ACL scripts need careful setup; a wrong condition can expose records or lock out valid users. • Depends on the availability of the ServiceNow instance.	5. AVAILABLE SOLUTIONS AS • Basic role-based ACLs: easy to set up, but they cannot check the Branch field value. • Manual record sharing / spreadsheets: no real-time control and a high risk of data leakage. • UI policies / client-side hiding: simple, but records can still be reached through lists or URLs. • Separate tables per branch: secure, but hard to maintain and duplicates data.
2. JOBS-TO-BE-DONE / PROBLEMS J&P • Restrict who can Read, Create, Write and Delete records in the Institution Details table. • Allow users with bb1 to read only EEE branch records. • Give separate permissions for Create (bb2), Write (bb3) and Delete (bb4). • Let administrators keep full access to all records.	9. PROBLEM ROOT CAUSE RC • The table holds ECE, EEE and CSE records, and plain table access exposes all of them. • A role-only ACL cannot filter records by a field value such as Branch. • Student information is sensitive and must be protected from unauthorized viewing or modification. • Hence a script-controlled ACL that checks both role and Branch is needed.	7. BEHAVIOUR BE • Directly related: admin creates users, assigns roles, builds the table and configures ACLs. • Admin impersonates test users to check Read, Create, Write and Delete results. • Users open lists and forms, and request role access from the administrator when blocked. • Indirectly associated: teams review and verify access permissions regularly.
3. TRIGGERS TR • Concern that students' records are visible to users from other branches. • New users or new branch records being added to the table. • Institution data-protection or security policy requiring controlled access. • An unexpected change or access to a record noticed by an administrator.	10. YOUR SOLUTION SL • Script-controlled ACLs on the custom table u_institution_details. • Read: role bb1 plus a script condition that allows access only when Branch = EEE. • Create, Write and Delete: restricted to roles bb2, bb3 and bb4 respectively. • Administrators retain full access to all records. • Behaviour is verified by testing each operation with different users, so authorized users get access and others are denied. • Fits the customer's need for secure, role-based and branch-based record control without any external integration.	8.1 ONLINE CH • ServiceNow lists and forms for viewing and managing Institution Details records. • System Security > Access Control (ACL) for configuring rules and scripts. • User impersonation to test access online.
4. EMOTIONS: BEFORE / AFTER EM • Before: concerned, insecure and worried about unauthorized access to student data. • After: confident, secure and in control, because every operation is checked by the ACL.		8.2 OFFLINE CH • Team discussion and requirement checking with the institution administrator. • Requests for role access raised verbally or by email. • Documented test results and checklists shared with the team.